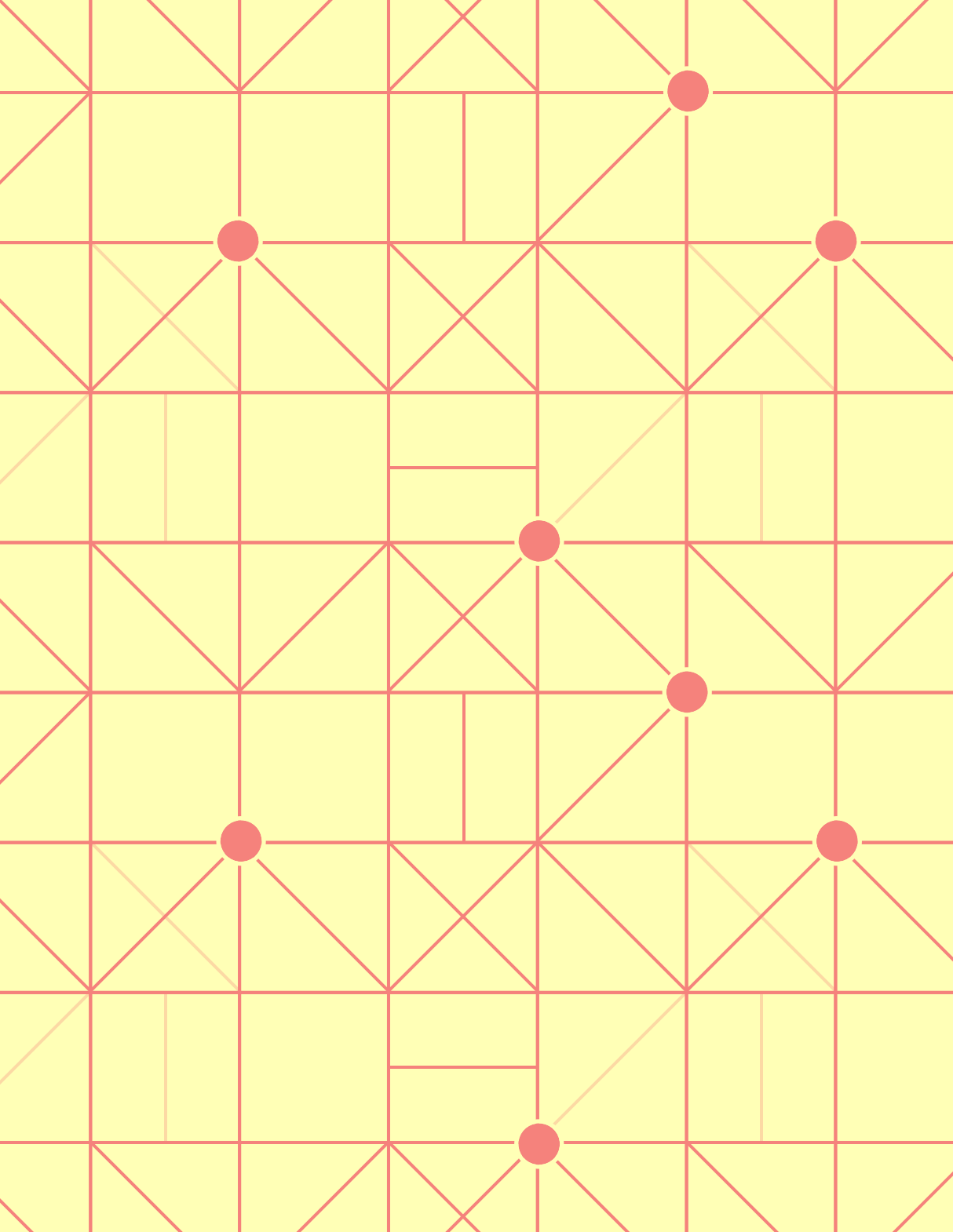




WORDS FROM LAST WEEK

SCHWIERIG - HARD

BRILLE - GLASSES

FEDERBALL - BADMINTON

STECKDOS - SOCKET

WACKEL KONTAKT - LOOSE CONTACT

ZUG - TRAIN

WALD - FOREST

DOCH - WELL, YES

ASCHENPUTTEL - CINDERELLA

FAHRRAD - BIKE

ENTE - DUCK

DUCKEN - DUCK

REGENSCHIRM - UMBRELLA

PFERD - HORSE

STRASSENVERKEHRSORDNUNG - TRAFFIC RULES

STIFT - PEN

FUSSBALL - FOOTBALL

HERBST - AUTUMN

TAUCHEN - DIVE

FLASCHE - BOTTLE

BRIEFTASCHE - WALLET

WYNDOR GLASS CO. & SIMPLEX METHOD

$$\begin{aligned}\max \quad Z &= 3x_1 + 5x_2 \\ x_1 &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \\ x_1 &\geq 0 \quad x_2 &\geq 0\end{aligned}$$

EX 4.4-4

$$\begin{aligned}\max \quad Z &= 2x_1 + 3x_2 \\ x_1 + 2x_2 &\leq 30 \\ x_1 + x_2 &\leq 20 \\ x_1 \geq 0 \quad x_2 \geq 0\end{aligned}$$

FIND THE OPTIMAL SOLUTION with SIMPLEX METHOD

- GEOMETRICALLY
- ALGEBRAICALLY

EX 4.4-5

$$\begin{aligned}\max \quad z &= 5x_1 + 9x_2 + 7x_3 \\ x_1 + 3x_2 + 2x_3 &\leq 10 \\ 3x_1 + 4x_2 + 2x_3 &\leq 12 \\ 2x_1 + x_2 + 2x_3 &\leq 8 \\ x_1, x_2, x_3 &\geq 0\end{aligned}$$

FIND THE SOL IN ALGEBRAIC FORM

EX 4.5 - 4

$$\begin{aligned}\max \quad Z &= 5x_1 + x_2 + 3x_3 + 4x_4 \\ x_1 - 2x_2 + 6x_3 + 3x_4 &\leq 20 \\ -4x_1 + 6x_2 + 5x_3 - 4x_4 &\leq 40 \\ 2x_1 - 3x_2 + 3x_3 + 8x_4 &\leq 50 \\ x_i &\geq 0 \quad i=1,2,3,4\end{aligned}$$

WORK THROUGH THE SIMPLEX METHOD TO DEMONSTRATE
THAT Z IS UNBOUNDED

EX 4.5-8

$$\begin{aligned} \max \quad Z &= 50x_1 + 25x_2 + 20x_3 + 40x_4 \\ &\quad 2x_1 + x_2 \leq 30 \\ &\quad x_3 + 2x_4 \leq 20 \\ x_j &\geq 0 \quad j=1-4 \end{aligned}$$

WORK THROUGH THE SIMPLEX METHOD TO FIND
ALL OPTIMAL SOLUTIONS